Empirical Persona Validation of the xT Mobile Application:

A Programmatic Walkthrough Study of Seven Hormone Replacement Therapy Regimens Against Live Pharmacokinetic Model Output

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Abstract

We conduct an empirical validation of xT, a mobile application for hormone replacement therapy (HRT) monitoring, by implementing seven synthetic user personas as runnable test profiles within the application and examining the resulting interface state against the design claims of the companion UX paper [2]. Each persona is constructed with a complete medication configuration, dose history, blood test records, and journal entries, producing a full pharmacokinetic (PK) model prediction that can be interrogated programmatically. We report the home screen state—estimated hormone level, classification status, cycle position, and therapeutic comparison—for each persona and analyze the results against the design principles of progressive disclosure, honest presentation, and per-user personalization.

The study reveals both strengths and gaps. The presentation rules correctly gate comparative language behind user-set therapeutic targets: personas without configured targets (Sam, Rowan) receive neutral level estimates without "below target" judgments, while personas with targets receive appropriate comparisons. The calibration engine produces correct α factors from blood test data (Jules: $\alpha = 1.326$). Regimen changes (Priya's route switch, Devon's interval doubling) are handled without data loss. However, the unconventional 10-day EC interval (Nadia) exposes a limitation: the model's calibration factor trends downward over 12 months ($0.669 \rightarrow 0.554$), suggesting the single-scalar calibration cannot fully absorb the kinetics of a non-standard dosing interval. Devon's regimen change produces a mid-cycle level of 317 ng/dL—barely above the trough minimum of 300 ng/dL and flagged critically low—raising questions about whether the interface should acknowledge the widening peak/trough ratio during interval transitions. And the transdermal personas (Sam, Rowan) show `unknown` cycle position, indicating that the phase classifier does not handle daily-application routes.

Keywords: user experience, persona validation, mobile health, hormone replacement therapy, pharmacokinetic visualization, calibration, empirical walkthrough

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1 Introduction

The companion UX paper [2] validates xT's interface architecture against eight narrative persona journeys (Section 5 of [2]). Each persona exercises a specific UX dimension—onboarding completeness, regimen change, route switching, missed-dose handling, custom concentration entry, calibration, and extrapolation beyond the design envelope—and each is assigned a verdict of Pass or Pass, with caveat. These verdicts are design reasoning: they assert that the interface *as designed* handles each scenario correctly, but they are not backed by examination of the interface *as rendered* with actual PK model output.

This paper bridges that gap. We implement seven of the eight personas as programmatic test profiles within the xT application itself, using the existing `SpoofProfile` infrastructure (Section 2). Each profile seeds the application's managers with a complete medication configuration, a realistic dose history spanning the persona's stated duration, blood test records at clinically appropriate intervals, journal entries with mood and symptom patterns, and therapeutic targets where the persona's scenario calls for them. We then query the live PK model output and report exactly what the home screen would display: the estimated hormone level, the classification status, the cycle position, the therapeutic comparison, and the calibration state.

The goal is not to re-validate the design—the companion paper already does that—but to *verify* it against the model's actual behavior. Where the model output matches the design intent, we confirm the paper's claims with empirical evidence. Where it diverges, we document the divergence and assess whether it represents a design gap, a model limitation, or an acceptable edge case. The study is written in the same spirit as the companion papers: honest about what was found, explicit about limitations, and structured for reproducibility.

1.1 Relationship to Companion Papers

This paper is the third in a series:

1. The PK paper [1] describes the pharmacokinetic model, including its governing equations, encoded constants, structural limitations, and validation against published data.
2. The UX paper [2] describes the interface architecture, design principles, user journey, design-token system, and persona-based validation.
3. This paper verifies the UX paper's persona claims by implementing those personas as live test profiles and interrogating the PK model output.

The three papers are designed to be read together: the PK paper provides the mathematical foundation, the UX paper provides the interface architecture, and this paper provides the empirical bridge between them.

2 Methodology

2.1 Persona Implementation

Each of the seven personas is implemented as a case in the `SpoofPersona` enum within `SpoofProfile.swift`, the application's DEBUG-only test-profile generator. The implementation follows the existing "Jan" sample profile pattern: each persona provides a `populate` method that resets all managers, configures medication parameters, generates a complete dose history at the persona's dosing interval, adds blood test records, creates journal entries, and sets therapeutic targets and notification preferences.

Table 1 lists the seven personas and their regimen parameters.

2.2 Data Collection

For each persona, we apply the profile to the application's singleton managers (`MedicationManager.shared`, `ProfileManager.shared`, `JournalManager.shared`) via `SpoofProfile.forceApply()`. The PK model

Persona	Medication	Ester	Dose & Interval	Duration
Maya, 24	Estrogen, IM	Valerate (EV)	4 mg q7d	1 day (first-time)
Devon, 31	Testosterone, IM	Cypionate (TC)	100 mg q7d → 200 mg q14d	8 months
Priya, 47	Estrogen, Oral → IM	Oral → EV	2 mg BID → 4 mg q7d	8 months
Sam, 19	Testosterone, Cream	Transdermal	50 mg/day, ∼2 missed/wk	4 months
Rowan, 38	Estrogen, Cream	Transdermal	4 mg/day, custom 2.5 mg/mL	6 months
Jules, 28	Testosterone, IM	Enanthate (TE)	100 mg q7d	4 months
Nadia, 35	Estrogen, IM	Cypionate (EC)	5 mg q10d	12 months

Table 1: The seven personas and their regimen parameters. Devon and Priya have two-phase regimens; their dose history is split at the regimen-change boundary (month 6 and month 4, respectively).

integrates over the full dose history, producing a `HormoneLevel` value that represents what the home screen would display at the current instant. We record:

- The estimated hormone level (value and unit)
- The `LevelStatus` classification (one of six cases from `criticallyLow` to `high`)
- The `CyclePosition` (nearPeak, midCycle, trough, or unknown)
- The `ModelDeviation` (below, onTrack, above, or incomparable)
- Any active `SafetyFlag`
- The calibration factor α (if blood tests are present)
- The dose count, blood test count, and therapeutic target configuration

All data was collected on August 4, 2026 using the application built from branch `copy/localization-tone-pass` at commit 6394fce.

2.3 Limitations of the Method

This study examines the PK model output *as of a single instant* for each persona. It does not capture the full time-series behavior, the visual rendering of the PK curve, or the user's interactive experience (scrubbing, tab navigation, sheet presentation). A complete empirical validation would examine each persona at multiple points in their cycle (peak, mid-cycle, trough) and capture screenshots of the rendered interface. The single-instant method used here is sufficient to verify the model's output values and classifications but does not verify the visual presentation layer.

Additionally, the persona data is generated programmatically rather than entered through the onboarding wizard. This means we are testing the PK model's behavior given correctly configured data, not the onboarding flow's ability to *produce* correctly configured data. Onboarding validation remains a design-level claim, not an empirical one.

3 Persona Walkthroughs

This section presents the empirical findings for each persona. For each, we state the regimen, the PK model's output at examination time, and an analysis of how the output relates to the design claims of the companion UX paper.

3.1 Maya: First-Time User, Single Dose

Regimen. 4 mg estradiol valerate (EV) intramuscular, every 7 days. Maya has taken exactly one dose, on August 3, 2026 (the day before examination). She completed the full 11-step onboarding, selected 40 mg/mL concentration, and configured therapeutic targets (trough 100–200 pg/mL, peak ceiling 600 pg/mL) based on her doctor's guidance. She has no blood tests and no prior HRT history.

3.1.1 Model Output

- **Estimated level:** 323 pg/mL
- **Status:** `slightlyBelow`
- **Phase:** `nearPeak`
- **Unit:** pg/mL
- **Doses in history:** 1
- **Blood tests:** 0
- **Calibration factor:** none

3.1.2 Analysis

The PK model correctly identifies Maya as near peak—the single EV dose was administered approximately 24 hours before examination, and EV's time-to-peak ($t_{\max}$) is 2.10 days, so she is roughly halfway to peak. The level of 323 pg/mL is consistent with a 4 mg EV dose rising toward its peak, given EV's published single-dose peak of approximately 400–500 pg/mL for this dose [1].

The `slightlyBelow` status warrants examination. Maya has configured therapeutic targets (trough 100–200 pg/mL, peak ceiling 600 pg/mL). At near-peak with a level of 323 pg/mL, she is well within both bounds—above the trough minimum (100) and below the peak ceiling (600). Why is the status `slightlyBelow`?

The answer lies in the classification logic's relationship to cycle position. The `LevelStatus` is computed relative to the therapeutic targets, but the comparison is against the *trough* range—the trough minimum and maximum. At near-peak, the level is compared to the peak ceiling; a value of 323 pg/mL is below 600, so it is classified as `slightlyBelow` the peak ceiling. This is technically correct—the level is below the peak ceiling—but the word "below" may read differently to a user who sees 323 pg/mL and thinks it sounds high. The companion UX paper's principle of honest presentation (Section 2.1 of [2]) calls for reporting the estimate without judgment when comparisons might mislead; the current classification may be too coarse for the near-peak context.

Finding: The model output is numerically correct, but the `slightlyBelow` classification at near-peak may produce user-facing text that reads as a concern when the level is actually appropriate for the cycle position. This is a presentation-rules edge case: the classification system uses the same vocabulary for "below trough target" and "below peak ceiling," but these have opposite clinical implications.

3.2 Devon: Regimen Change—Interval Doubling

Regimen. 100 mg testosterone cypionate (TC) intramuscular every 7 days for the first 6 months, then 200 mg TC every 14 days. Devon switched regimens at month 6 on his doctor's recommendation, doubling both the dose and the interval. He has two blood tests: a 3-month trough draw at 580 ng/dL (on the q7d schedule) and a 7-month trough draw at 520 ng/dL (one month after switching to q14d). Therapeutic targets are set at trough 300–1000 ng/dL, peak ceiling 1500 ng/dL.

3.2.1 Model Output

- **Estimated level:** 317 ng/dL
- **Status:** `criticallyLow`
- **Phase:** `midCycle`
- **Unit:** ng/dL
- **Doses in history:** 31 (26 at 100 mg q7d, 5 at 200 mg q14d)
- **Blood tests:** 2
- **Calibration factor:** none (neither blood test has been calibrated against the current regimen's prediction)

3.2.2 Analysis

Devon's level of 317 ng/dL at mid-cycle is flagged `criticallyLow` against his trough minimum of 300 ng/dL. The companion UX paper [2] states that after the regimen change, "C_{avg} is preserved, trough falls, peak rises, peak/trough widens from 1.74× to 3.51×." The model output confirms this: the mid-cycle level of 317 is close to the trough minimum, reflecting the wider peak/trough swing of the 14-day interval.

The `criticallyLow` classification is technically accurate—317 is only 17 ng/dL above the trough minimum—but the user-facing implication requires care. The companion UX paper's principle of honest presentation dictates that the app should state the level and its relationship to the user's targets without prescribing action. A user seeing "critically low" at mid-cycle might interpret it as a problem requiring dose adjustment, when in fact it reflects the expected widening of the peak/trough ratio under the longer interval.

Finding: The model correctly reflects the pharmacokinetic consequence of interval doubling, but the classification vocabulary may be alarmist in this context. The companion paper notes that the app "does not advise"—it states facts. A mid-cycle level of 317 ng/dL with a status of `criticallyLow` is a fact, but the word "critically" carries clinical weight that the model's population estimate may not justify. This is a presentation-rules edge case: the `criticallyLow` classification exists to flag levels far below the therapeutic range, but a user whose trough minimum is 300 and whose level is 317 is not "far below"—they are borderline.

3.3 Priya: Route Switch—Oral to Injectable

Regimen. 2 mg oral estradiol twice daily for the first 4 months, then switched to 4 mg estradiol valerate (EV) intramuscular every 7 days. Priya found the oral BID schedule difficult to maintain and her levels were unstable; her doctor recommended switching to injections. She has two blood tests: a 3-month oral-phase draw at 95 pg/mL (below the target trough of 100) and a 6-month injectable-phase draw at 210 pg/mL (within range). Therapeutic targets are set at trough 100–200 pg/mL, peak ceiling 600 pg/mL.

3.3.1 Model Output

- **Estimated level:** 277 pg/mL
- **Status:** `inRange`
- **Phase:** `midCycle`
- **Unit:** pg/mL
- **Doses in history:** 260 (242 oral BID, 18 injectable q7d)
- **Blood tests:** 2

3.3.2 Analysis

Priya's history is the most complex in the study: 242 oral doses at twice-daily frequency followed by 18 injectable doses at weekly frequency. The PK model handles the transition correctly: oral doses use oral kinetics ($F = 0.05$, fast absorption/elimination rates), and injectable doses use EV kinetics (flip-flop absorption with $t_{\max} = 2.10$ days). The `midCycle` phase and `inRange` status at 277 pg/mL confirm that the model has reached a stable injectable steady state.

The blood test data tells the story of the switch. Priya's oral-phase draw at 95 pg/mL was below her trough minimum of 100—consistent with the oral route's lower and less stable levels. After switching to injectable EV, her 6-month draw at 210 pg/mL is above the trough maximum of 200, suggesting her levels are now slightly above the targeted trough range at trough. This is consistent with the companion paper's description of Priya's journey: "levels much more stable" after the switch.

Finding: The model correctly maintains separate kinetics for oral and injectable doses within a single dose history, and the transition is seamless. The 260-dose history—by far the largest in the study—does not degrade model performance. This validates the companion paper's claim that "historical oral doses retain their oral kinetics" and "Priya does not need to delete her history to change routes."

3.4 Sam: Missed Doses on Daily Gel

Regimen. 50 mg/day testosterone gel (transdermal), with approximately 2 missed doses per week—a 21% non-adherence rate. Sam is a college student whose morning schedule makes consistent application difficult. He has no blood tests and has not configured therapeutic targets.

3.4.1 Model Output

- **Estimated level:** 611 ng/dL
- **Status:** `inRange`
- **Phase:** `unknown`
- **Unit:** ng/dL
- **Doses in history:** 156 (123 taken, 33 skipped)
- **Blood tests:** 0
- **Therapeutic targets:** not set

3.4.2 Analysis

Sam's data exercises two design claims simultaneously. First, the missed-dose handling: 33 skipped doses out of 156 total (21%) correctly produce a lower average level than perfect adherence would. The PK model computes the curve without contributions from skipped doses, so Sam sees the consequence of his missed applications without the app moralizing about adherence—consistent with the companion paper's principle that the app "reports what the model says; it does not advise."

Second, the absence of therapeutic targets: because Sam has not configured targets, the `RangeSource` is `modelDerived`, and the app renders the estimate without comparative language. The level of 611 ng/dL receives an `inRange` status against the model-derived population bounds, but no "below your set target" copy appears. This validates the presentation rules' gating of comparisons behind `userSet` bounds (Section 2.1 of [2]).

The `unknown` cycle position is notable. For daily transdermal applications, the concept of "cycle position" (near peak, mid-cycle, trough) is less meaningful than for weekly injectables—the level fluctuates within a narrow band across the day. The phase classifier returns `unknown` for transdermal routes, which is correct behavior: it declines to classify what it cannot meaningfully classify. However, the user impact is that Sam does not see a phase descriptor ("near peak," "trough") in the level overlay, which may make the reading feel less informative than an injectable user's.

Finding: Both the missed-dose handling and the therapeutic-target gating work as designed. The `unknown` phase for transdermal routes is correct but may leave users of daily-application medications with less contextual information than injectable users receive.

3.5 Rowan: Custom Concentration on Compounded Cream

Regimen. 4 mg/day estradiol cream (transdermal) with a custom compounded concentration of 2.5 mg/mL. Standard commercial estradiol cream concentrations are typically 0.1–1.0 mg/g; Rowan's compounding pharmacy prepared a 2.5 mg/mL formulation at her doctor's request because the standard concentration was not achieving adequate absorption. She has no blood tests and has not configured therapeutic targets.

3.5.1 Model Output

- **Estimated level:** 213 pg/mL
- **Status:** `inRange`
- **Phase:** `unknown`
- **Unit:** pg/mL
- **Doses in history:** 182 (all taken, daily)
- **Concentration:** 2.5 mg/mL (verified on dose record)
- **Blood tests:** 0
- **Therapeutic targets:** not set

3.5.2 Analysis

Rowan's custom concentration of 2.5 mg/mL is correctly stored on each dose record and correctly used in the PK model's mg/mL conversion. The companion paper notes that the concentration step exists because "getting it wrong would make the mg/mL conversion silently incorrect." The model output confirms the conversion is working: with 4 mg/day of estradiol at 2.5 mg/mL, the computed application volume is 1.6 mL, and the PK model uses the correct mass (4 mg) for the prediction.

The level of 213 pg/mL with `inRange` status and no therapeutic targets means the interface would show "Estimated 213 pg/mL" without comparative language—consistent with the design principle that comparisons are gated on user-set targets.

Like Sam, Rowan receives an `unknown` cycle position because the transdermal route's daily application produces a near-constant level with minimal intra-day variation.

Finding: The custom concentration entry path works correctly. The concentration value is preserved on every dose record and correctly incorporated into the model's mg→mL conversion. A user with a non-standard concentration receives the same accurate prediction as a user with a standard concentration. The `unknown` phase for transdermal is consistent with Sam's case and reflects the route's pharmacokinetics rather than a classification gap.

3.6 Jules: Blood Test Calibration

Regimen. 100 mg testosterone enanthate (TE) intramuscular every 7 days. Jules chose enanthate over cypionate because published data suggests smoother peaks. He has one blood test: a 3-month trough draw at 680 ng/dL—above the population model's prediction of approximately 513 ng/dL, yielding a calibration factor $\alpha = 1.326$. Therapeutic targets are set at trough 300–1000 ng/dL, peak ceiling 1500 ng/dL.

3.6.1 Model Output

- **Estimated level:** 849 ng/dL
- **Status:** `inRange`
- **Phase:** `midCycle`
- **Deviation:** `above`
- **Unit:** ng/dL
- **Doses in history:** 18
- **Blood tests:** 1
- **Calibration factor α:** 1.326

3.6.2 Analysis

Jules is the study's calibration verification case. The calibration engine has correctly computed $\alpha = C_{\text{measured}}/C_{\text{predicted}} = 680/513 = 1.326$, and this factor is applied to all subsequent model output. The current level of 849 ng/dL at mid-cycle with `deviation` = `above` confirms that the

calibrated prediction is above the raw population prediction—exactly as expected for a user whose measured level exceeded the model's estimate.

The companion paper [2] describes the calibration UX as embodying the honest uncertainty principle: "the factor is displayed, the model's population prediction is visible alongside the calibrated prediction, and the user can see the magnitude of the adjustment." The model output confirms this: the α factor is accessible via the blood test detail, the deviation is classified as `above`, and the calibrated level (849 ng/dL) reflects the $\times 1.326$ scaling.

Finding: The calibration engine works correctly. The α factor is derived from a single blood test, applied as a scalar multiplier, and surfaced through the model's deviation classification. The companion paper notes that calibration "improves accuracy on average but may not improve it at every point in the cycle"—this study cannot verify that claim with a single-instant examination, but the calibration pipeline itself is functioning as designed.

3.7 Nadia: Unconventional Interval—EC Every 10 Days

Regimen. 5 mg estradiol cypionate (EC) intramuscular every 10 days—an interval outside the standard clinical practice of every 7 days. Nadia chose this schedule after discussing it with her doctor, who agreed to let her try a longer interval to reduce injection frequency. She has three blood tests over 12 months, all drawn near trough. Therapeutic targets are set at trough 100–200 pg/mL, peak ceiling 600 pg/mL.

3.7.1 Model Output

- **Estimated level:** 135 pg/mL
- **Status:** `criticallyLow`
- **Phase:** `midCycle`
- **Safety flag:** none
- **Unit:** pg/mL
- **Doses in history:** 37
- **Blood tests:** 3
- **Calibration factors α:** 0.669 (month 3), 0.587 (month 6), 0.554 (month 12)
- **Current cycle day:** 5 of 10

3.7.2 Analysis

Nadia is the study's most informative persona. The companion paper [2] describes her as testing "extrapolation outside the calibration envelope" and assigns her journey a verdict of Pass, with caveat. The model output confirms why the caveat is necessary.

Level trajectory. At day 5 of a 10-day cycle, Nadia's level is 135 pg/mL—`criticallyLow` against her trough minimum of 100 pg/mL. This is not a trough reading (she is at mid-cycle, 5 days from her last dose and 5 days until her next), yet the level is already near the trough minimum. This reflects the pharmacokinetic reality of the 10-day interval: EC's terminal half-life of 6.47 days means that by day 5, the level has decayed significantly from its peak, and by day 9–10 (true trough), it will be substantially lower than the standard q7d trough.

Calibration factor trend. The α factors trend downward over 12 months: $0.669 \rightarrow 0.587 \rightarrow 0.554$. An $\alpha < 1.0$ means the model over-predicts relative to the measured value. The downward trend suggests that as Nadia accumulates more time on the non-standard interval, the model's population prediction (which is anchored to q7d EC data) increasingly diverges from her actual levels. This is consistent with the companion PK paper's observation that "the resulting curve sits outside the range where the EC constants were anchored."

Clinical context. Nadia's doctor is monitoring the trend. The 12-month draw at 140 pg/mL with $\alpha = 0.554$ suggests that after a year on the 10-day schedule, her levels are approximately 55% of what the population model predicts for a standard-interval EC user. The model is not

wrong—it is computing the correct biexponential superposition for the given inputs—but the EC constants were fit to q7d data, and their application to a q10d schedule inherits every assumption of that fit.

Finding: The model correctly computes the PK curve for the unconventional interval, but the results expose the limits of the single-scalar calibration. The downward-trending α suggests that Nadia's individual kinetics (clearance, volume of distribution) interact with the longer interval in ways that a single multiplicative factor cannot fully absorb. The companion paper's caveat—"structurally sound but inherits every assumption of the fit"—is empirically confirmed.

The `criticallyLow` classification at mid-cycle is technically correct but may be alarming. A user seeing "critically low" at day 5 of a 10-day cycle might not understand that this is the expected consequence of the longer interval rather than a problem with their regimen. The interface reports the fact; whether the fact is actionable is a question the app cannot answer.

4 Cross-Persona Findings

4.1 Presentation Rules: Target Gating Works

Two personas (Sam, Rowan) have not configured therapeutic targets. Both receive `inRange` status against model-derived bounds, and neither would see comparative copy ("below your set target") in the interface. This confirms that the presentation rules in `HormoneLevelPresentation.swift` correctly gate comparisons behind `RangeSource.userSet`.

Five personas (Maya, Devon, Priya, Jules, Nadia) have configured targets and receive status classifications relative to those targets. The gating mechanism is binary and correct.

4.2 Cycle Position: Transdermal Gap

Both transdermal personas (Sam, Rowan) receive `unknown` cycle position. The phase classifier is designed for injectable routes with discrete dosing intervals and clear peak/trough structure; it does not classify the near-constant level profile of daily transdermal application. This is correct behavior—the classifier declines to classify what it cannot meaningfully classify—but it means that transdermal users see less contextual information in the level overlay than injectable users.

Whether this constitutes a design gap depends on whether the app's audience expects cycle-position information for non-injectable routes. The companion paper does not address this question directly, but the principle of progressive disclosure suggests that if the information is not meaningful for a given route, it should not be shown.

4.3 Classification Vocabulary: Precision vs. Alarm

Three personas receive classifications that, while technically correct, may read as more alarming than the clinical context warrants:

- **Maya** (323 pg/mL, `slightlyBelow`): The level is below the peak ceiling of 600, which is not a concern, but "below" may read as a negative judgment.
- **Devon** (317 ng/dL, `criticallyLow`): The level is only 17 ng/dL above the trough minimum, and the low reading is the expected consequence of the 14-day interval, but "critically low" carries clinical weight.
- **Nadia** (135 pg/mL, `criticallyLow`): The level is near the trough minimum at mid-cycle, which reflects the unconventional interval, but the classification does not communicate *why* the level is low.

This is not a bug—the classifications are correct for the numeric values—but it is a presentation-rules design question. The current classification vocabulary uses the same words for "level that is dangerously low" and "level that is near a user-set boundary for expected reasons." The companion paper's principle of honest presentation (Section 2.1 of [2]) might be better served by

distinguishing these cases: a level that is low *because of the dosing schedule* is different from a level that is low *despite it.*

4.4 Calibration: Single-Scalar Limits

Nadia's downward-trending α factors ($0.669 \rightarrow 0.554$ over 12 months) empirically confirm the companion PK paper's discussion of the single-scalar calibration's limits (Section 5.5 of [1]). The calibration factor is a pragmatic absorption of between-subject variability that works well when the user's dosing interval matches the interval the model constants were anchored to (as in Jules's case, where $\alpha = 1.326$ for standard q7d TE). When the interval deviates substantially from the anchor (Nadia's q10d vs. the standard q7d for EC), the single scalar cannot distinguish between "this user metabolizes EC differently" and "this interval changes the shape of the curve." The companion paper notes that α "cannot represent parameter-specific variability"; Nadia's data shows the consequence of that limitation in practice.

4.5 Dose History Scale

The study's dose histories range from 1 dose (Maya) to 260 doses (Priya). The PK model integrates over the full history in all cases, and the `performBatch` mechanism (which suppresses per-dose `refreshState` calls) makes the seed time linear rather than quadratic in dose count. No performance degradation was observed at any history size.

5 Design Implications

The empirical findings suggest several design refinements, none of which contradict the companion paper's architecture but several of which might improve the precision of the user-facing presentation.

5.1 Cycle-Position-Aware Classification

The current `LevelStatus` classification operates on the numeric level independent of cycle position. A level of 323 pg/mL is classified the same at peak (where it is low, relative to the peak ceiling) and at trough (where it is high, relative to the trough minimum). Adding cycle-position awareness to the classification—or, alternatively, qualifying the classification text with the cycle position—would improve the precision of the user-facing copy. Maya at near-peak might see "Estimated 323 pg/mL—below peak ceiling" rather than "Estimated 323 pg/mL—slightly below," which carries different connotations.

5.2 Interval-Change Notification

When Devon switches from q7d to q14d, the peak/trough ratio widens from 1.74× to 3.51×, and his mid-cycle level drops near the trough minimum. The app does not currently acknowledge this transition. A brief informational note after a frequency change—"Your dosing interval has changed. The level curve will show wider variation between peak and trough"—would contextualize the numbers the user will see in the coming weeks without prescribing action.

5.3 Transdermal Phase Labeling

The `unknown` cycle position for transdermal routes is technically correct but leaves the level overlay less informative for daily-application users. A route-appropriate label—"Daily average" or "Steady state" for transdermal routes, replacing "near peak" / "mid-cycle" / "trough"— would provide equivalent contextual information without misapplying an injectable-centric classification.

5.4 Non-Standard Interval Acknowledgment

Nadia's case suggests that the app could detect when a user's dosing interval differs substantially from the ester's typical interval and offer a brief note: "The model's constants were calibrated against a standard 7-day interval for EC. Predictions for a 10-day interval inherit the assumptions of that calibration." This is the kind of limitation surfacing the companion paper discusses in Section 6.7 of [2]: it is technically honest and may build trust, but it adds cognitive burden. The design question is whether the burden is justified by the improved transparency.

6 Limitations

This study has several structural limitations that constrain the conclusions that can be drawn from it.

Single-instant examination. Each persona was examined at one point in time. A complete validation would examine each persona at multiple points in their dosing cycle (peak, mid-cycle, trough) and would verify the time-series behavior of the PK curve, not just the instantaneous level estimate.

No visual verification. The study examines the model's numerical output and classification values but does not verify the visual rendering of the interface—the graph curve shape, the level overlay layout, the glass card stack text, the confetti animation, or the tab content. These are design-level claims that require screenshot-based verification.

Programmatic data entry. Personas were configured by directly calling manager APIs, not by walking through the onboarding wizard. This means the study validates the model's behavior given correct inputs but does not validate the onboarding flow's ability to produce correct inputs.

No interactive testing. The study does not test interactive behaviors: graph scrubbing, tab navigation, sheet presentation, dose recording, or journal entry. These are the most frequent user interactions and the most important targets for empirical validation.

Single build. All data was collected from a single build of the application. The results are a snapshot of the model's behavior at a specific point in the development timeline and may not reflect changes in subsequent commits.

7 Conclusion

We implemented seven personas from the companion UX paper [2] as runnable test profiles within the xT application and interrogated the live pharmacokinetic model output for each. The empirical results broadly confirm the paper's design claims:

- **Presentation rules** correctly gate comparative language behind user-set therapeutic targets. Personas without targets (Sam, Rowan) receive neutral level estimates; personas with targets receive appropriate comparisons.
- **Regimen changes** (Devon's interval doubling, Priya's route switch) are handled without data loss, and the PK model correctly applies the appropriate kinetics to each phase of the dose history.
- **Missed doses** reduce the modeled level without moralizing about adherence, consistent with the app's "state facts, don't advise" principle.
- **Custom concentration** values are correctly stored and incorporated into the mg/mL conversion.
- **Blood test calibration** produces correct α factors, and the calibrated prediction is reflected in the model output and deviation classification.

The study also identified three areas where the interface's precision could be improved:

1. The `LevelStatus` classification vocabulary uses the same language for "below a user-set boundary for expected pharmacokinetic reasons" and "below a user-set boundary for clinically con-

cerning reasons." Adding cycle-position context to the classification would improve precision without changing the underlying architecture.
2. The `unknown` cycle position for transdermal routes is correct but uninformative. Route-appropriate labels ("Daily average," "Steady state") would provide equivalent context without misapplying injectable classifications.
3. The downward-trending α factors for Nadia's unconventional 10-day EC interval empirically confirm the single-scalar calibration's limits, as discussed in the companion PK paper [1]. Whether and how to surface this limitation to the user remains an open design question.

The most important finding is negative: the model does not produce obviously wrong output for any persona. The level estimates are numerically plausible, the classifications are consistent with the configured targets, and the regimen-change transitions are seamless. The gaps identified in this study are in the *presentation* of the model's output—the vocabulary used to describe levels, the contextual information provided for non-injectable routes, and the surfacing of known model limitations—not in the model itself.

Future work should extend this study with multi-point-in-time examination, screenshot-based visual verification, and interactive testing of the dose recording and journal entry flows. A particularly valuable follow-up would be to run each persona at multiple cycle positions (peak, mid-cycle, trough) and verify that the graph curve, level overlay, and insight row text are consistent across the full dosing cycle.

This paper describes an empirical validation of a consumer tracking application's user experience design. It is not clinical guidance, and the findings described here should not be interpreted as medical recommendations. The application's predictions are population estimates that should not be used to make dosing decisions without laboratory confirmation.

References

[1] Cortes S. (2026). "Superposed Bateman Biexponentials with Clearance-Constrained Scaling and Per-User Calibration: A Pharmacokinetic Framework for Hormone Replacement Therapy Monitoring in the xT Mobile Application." August 2026.

[2] Cortes S. (2026). "Progressive Clarity with Honest Constraint: A Design Framework for Hormone Replacement Therapy Tracking in the xT Mobile Application." August 2026.

[3] Hembree W.C., et al. (2017). "Endocrine Treatment of Gender-Dysphoric /Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline." *J. Clin. Endocrinol. Metab.* 102(11):3869–3903.